

JEFFREY SNOWDEN

jeffreysnowden@gmail.com

PROFESSIONAL EXPERIENCE

Research Scientist

BSCS Science Learning – 2023 to Present

Research Associate

BSCS Science Learning – 2018 to 2023

Application Support Specialist

BSCS Science Learning – 2016 to 2018

Clerical Assistant

BSCS Science Learning – 2015 to 2016

EDUCATION

M.A., Research and Evaluation Methods, University of Colorado, Denver. 2020

B.A., Philosophy, University of Colorado, Colorado Springs. 2016

B.A., Communication, University of Colorado, Colorado Springs. 2016

A.A., Philosophy, Pikes Peak Community College. 2014

PEER-REVIEWED ARTICLES AND CHAPTERS

Donovan, B. M., Weindling, M., Amemiya, J., Salazar, B., Lee, D., Syed, A., ... & **Snowden, J.** (2024). Humane genomics education can reduce racism. *Science*, 383(6685), 818-822.

Harris, E., Lowe, I., Mohan, L., Cohen, W., Severance, S., Giotto, T., ... & **Snowden, J.** (2024). Designing Trellises: Cultivating science and engineering in the garden. *Science and Children*, 61(2), 70-75.

Herrmann-Abell, C. F., Stuhlsatz, M. A., Wilson, C. D., **Snowden, J.**, & Donovan, B. M. (2023). Using Rasch Measurement to Develop Model-Based Reasoning Assessment Tasks. In *Advances in Applications of Rasch Measurement in Science Education* (pp. 241-264). Cham: Springer International Publishing.

Schultheis, E. H., Kjolvik, M. K., **Snowden, J.**, Mead, L., & Stuhlsatz, M. A. (2022). Effects of Data Nuggets on Student Interest in STEM Careers, Self-efficacy in Data Tasks, and

PEER-REVIEWED CONFERENCE PROCEEDINGS

Paper Presentations

Snowden, J., Donovan, B., Mohan, L., & Harris, E. (2025, April). *The Impact of Localized Climate Change Curriculum on Student Environmental Science Agency*. [Conference paper]. American Educational Research Association, Denver, CO.

Snowden, J., Harris, E., & Mohan, L. (2025, March). *Localizing climate change education: Impacts on student knowledge and agency in high school science classrooms*. [Conference paper]. National Association for Research in Science Teaching, National Harbor, MD.

Guy-Gaytán, C., **Snowden, J.**, Mohan, L. & Harris, E. (2025, March). *Differences between students and teachers in the perceived relevance of a localized climate change unit*. [Round table paper]. National Association for Research in Science Teaching, National Harbor, MD.

Passmore, C., Ghanimi, H., Herrmann-Abell, C., Olson, P., **Snowden, J.**, Stuhlsatz, M., & Wilson, C. (2025, March). *Variations in Epistemological Messaging in High School Biology*. [Conference paper]. National Association for Research in Science Teaching, National Harbor, MD.

Mendez, S. L., Tygret, J. A., Bruwer, A., Stuhlsatz, M., & **Snowden, J.** (2024, June). *Improving university postdoctoral affairs offices: Viewpoints from engineering postdoctoral scholars of color*. Paper presented at the American Society for Engineering Education Annual Conference & Exposition, Portland, OR.

Snowden, J., Donovan, B., Mohan, L., Harris, E. (2024). *Evaluating Student Engagement in Climate Change Education: A Novel Approach to Measuring Environmental Science Agency*. Paper presentation at the National Association for Research in Science Teaching (NARST) Annual International Conference, Denver, CO.

Wilson, C., Passmore, C., Stuhlsatz, M., Abell, C., **Snowden, J.**, Ghanimi, H. & Olson, P. (2024). *Impacts & Moderation of a Model-Based High School Biology Program on Student Outcomes*. Paper presentation at the National Association for Research in Science Teaching (NARST) Annual International Conference, Denver, CO.

Herrmann-Abell, C., **Snowden, J.**, Stuhlsatz, M., Donovan, B., Passmore, C., Olson, P. & Wilson, D. (2024) *Measuring the Impact of Multidimensional Science Instructional Materials*. Paper presented at the American Educational Research Association (AERA) Conference, Philadelphia, PA.

- Snowden, J.,** Stuhlsatz, M., & Mendez, S. (2022). *Development of the Process Documentation Tool to Assess Model Development*. Paper presentation at the Alliance for Graduate Education and the Professoriate (AGEP) National Research Conference, Corpus Christi, TX.
- Hermann-Abell, C., Donovan, B., Harris, E., **Snowden, J.,** Stuhlsatz, M., & Wilson, D. (2022) *Developing Model-Based Reasoning Assessment Tasks to Investigate the Efficacy of High School Biology Instructional Materials*. Paper presented at the American Educational Research Association (AERA) Conference, San Diego, CA.
- Kowalski, S., **Snowden, J.,** & Carey, L. (2021) *A Quasi-experimental Study of the Efficacy of a Designed-for-NGSS Unit and PD*. Paper presentation at the annual meeting of the National Association of Research on Science Teaching (NARST), Virtual.
- Stuhlsatz, M., Kjervik, M., Schultheis, E., **Snowden, J.,** Donovan, B., & Mead, L., (2021). *Measuring the Efficacy of an Approach to Integrating Quantitative Reasoning in High School Biology*. Paper presentation at the annual meeting of the National Association of Research on Science Teaching (NARST), Virtual.
- Kowalski, S., Mohan, L., Stennett, B., Stimac, C., & **Snowden, J.** (2020). *Developing a unit designed for NGSS: Successes and lessons learned in the development process*. Paper set at the annual meeting of the National Association of Research on Science Teaching (NARST), Portland, OR.
- Kowalski, S., Mohan, L., Stennett, B., Stimac, C., & **Snowden, J.** (2020). *Developing a Media-Rich Digital Unit to Support 3D Teaching and Learning Subject/Problem*. Paper set at the annual meeting of the National Association of Research on Science Teaching (NARST), Portland, OR.
- Kowalski, S., Mohan, L., Stennett, B., Stimac, C., & **Snowden, J.** (2020). *Professional Development for A Medical Mystery: Moving Beyond the Curriculum*. Paper set at the annual meeting of the National Association of Research on Science Teaching (NARST), Portland, OR.
- Kowalski, S., Mohan, L., Stennett, B., Stimac, C., & **Snowden, J.** (2020). *A Quasi-experimental Study of the Efficacy of a Designed-for-NGSS Unit and PD Subject/Problem*. Paper set at the annual meeting of the National Association of Research on Science Teaching (NARST), Portland, OR.
- Stuhlsatz, M., Mead, L., Schultheis, E., Kjervik, M., & **Snowden, J.** (2020). *Measuring the Efficacy of an Approach to Integrating Quantitative Reasoning in High School Biology*. Paper presentation at the annual meeting of the National Association of Research on Science Teaching (NARST), Portland, OR.

Poster Presentations

- Deverel-Rico, C., Herrmann-Abell, C., **Snowden, J.**, Wilson, C., & Olson, P. (2025, October). *Is there an 'experience gap' with how students experience middle school science assessments?* [Conference poster]. Society for Research on Educational Effectiveness, Chicago, IL.
- Snowden, J.**, Mohan, L., Harris, E., Guy-Gaytan, C., Mohan, A., & Lee, D. (2025, June). *Localized Climate Education: Examining Impacts on Students' Environmental Science Agency.* [Conference poster]. DRK-12 PI Meeting, Arlington, VA.
- Mohan, L., Harris, E., Guy-Gaytán, C., & **Snowden, J.** (2025, March). *Impacts of professional learning to support teachers' design capacity for localized climate units.* [Poster presentation]. National Association for Research in Science Teaching, National Harbor, MD.
- Snowden, J.**, Stuhlsatz, M., Mendez, S., & Bruwer, A. (2024, October). *The Secret Sauce: The Impact of the NSG AGEP Engineering Alliance PRIMED on Engineering Postdoctoral Scholars.* Poster presented at the NSF AGEP@25 Conference: Celebrating the Past, Creating the Future Conference, Crystal City, VA.
- Abell, C., **Snowden, J.**, Stuhlsatz, M., Donovan, B., Passmore, C. & Wilson, C. (2024, March 17 – 20). *Exploring the Impacts of Educative Model-Based Biology Instructional Materials on Teacher Outcomes.* [Poster presentation]. National Association for Research in Science Teaching (NARST) Annual International Conference. Denver, CO.
- Wilson, C., Abell, C., Bintz, J., Lo, A., Mohan, L., Flanagan, J., Guy-Gaytan, C., Rojas, D., **Snowden, J.**, Stennett, B., & Hsi, S. (2024, March 17 – 20). *BSCS Then and Now: Advancing High Quality Science Education for All Learners.* [Poster symposium]. National Association for Research in Science Teaching (NARST) Annual International Conference. Denver, CO.
- Passmore, C., Wilson, C., Stuhlsatz, M., Ghanimi, H., Abell, C., **Snowden, J.**, Donovan, B., & Olson, P. (2023, June 28 – 30). *Development and Use of a Remote Observation Tool for Examining Impacts of the Model based Biology Educational Resource (MBER) on Instructional Practice.* [Poster presentation]. DRK-12 PI Meeting. Arlington, VA.
- Wilson, C., Passmore, C., Stuhlsatz, M., Abell, C., **Snowden, J.**, Ghanimi, H., Donovan, B., & Olson, P. (2023, June 28 – 30). *Impacts & Moderation of the Model-Based Biology Education Resource (MBER) on Student Outcomes.* [Poster presentation]. DRK-12 PI Meeting. Arlington, VA.
- Snowden J.**, Kowalski, S., Askinas, K., Weindling, M., & Stuhlsatz, M. (2023, June 28 – 30). *Teaching Students About Variation and Covariation: What Do We Know, and What Do We Need to Find Out?* [Structured poster presentation]. DRK-12 PI Meeting. Arlington, VA.

- Snowden, J.**, Mohan, L., Harris, E., Donovan, B., Carey, L., Dixon, C., Guy-Gaytán, C., Mohan, A., Stennett, B., Stimac, C., & Young, H. (2023, June 28 – 30). *Supporting High School Students and Teachers with a Digital, Localizable, Climate Education Experience: Quantitative Instruments to Measure Students' Environmental Science Agency*. [Structured poster presentation]. DRK-12 PI Meeting. Arlington, VA.
- Kowalski, S., Belcastro, A., Hvidsten, C., Ollison, G., Constantine, A., Faruqi, F., Askinas, K., De Vul, R., **Snowden, J.**, & Roherig, G. (2021) *Design Decisions in Translating Face-to-Face Video-Based Elementary Science Professional Development to an Online Environment*. Poster presentation at the American Educational Research Association (AERA) Conference, Virtual.
- Stennett, B., **Snowden, J.**, Kowalski, S., Mohan, L., Carey, L., Young, H., & Stimac, C. (2021) *A Quasi-Experimental Study of Next Generation Science Standards Curriculum and Online Professional Development on Three-Dimensional Learning Outcomes*. Poster presentation at the American Educational Research Association (AERA) Conference, Virtual.
- Hermann-Abell, C., Donovan, B., Harris, E., **Snowden, J.**, Stuhlsatz, M., & Wilson, D. (2021) *Developing Assessment Tasks to Measure Model-Based Reasoning in Biology*. Poster presentation at the annual meeting of the National Association of Research on Science Teaching (NARST), Virtual.
- Harris, E., Mohan, L., Cohen, W., Severance, S., & **Snowden, J.** (2021). *Place-based Storyline Design: Selecting an Anchoring Problem for Engineering in the Garden*. Paper presentation at the annual meeting of the National Association of Research on Science Teaching (NARST), Virtual.
- Kowalski, S., **Snowden, J.**, Carey, L., Stennett, B., Mohan, L., Young, H., & Stimac, C. (2020) *A Quasi-experimental Study of NGSS Curriculum and PD on Three-dimensional Learning Outcomes*. Poster presentation at the Society for Research on Educational Effectiveness (SREE) Conference, Crystal City, VA.

Video

- Harris, E., Cohen, W., Giotta, J., Lowe, I., Severance, S., & **Snowden, J.** (2022). Engineering in the Garden. *National Science Foundation (NSF) STEM for All Video Showcase*.

CURRENT RESEARCH

- Collaborative Research: Extending and Investigating the Impact of the High School Model-Based Educational Resource* (2018 – 2024). PI: Christopher Wilson. National Science Foundation, Discovery Research K-12 (Award #1813538).
- Collaborative Research: The AGEP Engineering Alliance: A Model to Advance Historically Underrepresented Minority Postdoctoral Scholars and Early Career Faculty in*

Engineering (2018 – 2024). PI: Comas Haynes. National Science Foundation, Includes (Award #1821298).

Supporting High School Students and Teachers with a Digital, Localizable, Climate Education Experience (2021 – 2025). PI: Lindsey Mohan. National Science Foundation Collaborative Archive & Data Research Environment (CADRE) (Award # 2100808).

Towards a More Human(e) Genetics Education: Exploring how Knowledge of Genetic Variation and Causation Affects Racial Bias among Adolescents (2017 – 2024). PI: Brian Donovan. National Science Foundation, Division of Research On Learning (Award # 1660985).

PREVIOUS RESEARCH

Teacher Learning and Leadership Through Agricultural Engineering (TILLAGE) (2020 – 2022). PI: Emily Harris. National Institute of Food and Agriculture (Award #2020-67037-31047).

Teaching Students to Reason about Variation and Covariation in Data: What Do We Know and What Do We Need to Find Out? (2019 – 2022). PI: Susan Kowalski. Collaborative Archive & Data Research Environment (CADRE) (Award #1920119).

Translating a Video-based Model of Teacher Professional Development to an Online Environment (2018 – 2021). PI: Susan Kowalski. National Science Foundation, Discovery Research K-12 (Award # 1813127).

Collaborative Research: Scientific Data in Schools: Measuring the efficacy of an innovative approach to integrating quantitative reasoning in secondary science (2015 – 2020). PI: Molly Stuhlsatz. National Science Foundation, Discovery Research K-12 (Award #1503005).

Three-Dimensional Teaching and Learning: Rebuilding and Researching an Online Middle School Curriculum (2015 – 2020). PI: Susan Kowalski. National Science Foundation, Discovery Research K-12 (Award #1502571).

HONORS AND AWARDS

UCCS Degree Honors: Magna Cum Laude & With Highest Distinction

UCCS Presidents List: Fall 2016

UCCS Deans List: Fall 2014, Spring 2015, Fall 2015, Spring 2016

PPCC Presidents List: Spring 2013, Fall 2013, Spring 2014

MEMBERSHIPS

American Educational Research Association (AERA)

American Evaluation Association (AEA)

National Association for Research in Science Teaching (NARST)